

Orderbook Feature Learning and Asymmetric Generalization in Intraday Electricity Markets

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ABSTRACT

Accurate probabilistic forecasting of intraday electricity prices is critical for market participants to inform trading decisions. Existing studies rely on specific domain features, such as volume-weighted average price and the last price. However, the rich information in the orderbook remains underexplored. Furthermore, these approaches are often developed within a single country and product type, making it unclear whether the approaches are generalizable. In this paper, we extract 384 features from the orderbook and identify a set of powerful features via feature selection. Based on selected features, we present a comprehensive benchmark using classical statistical models, tree-based ensembles, and deep learning models across two countries and two product types. We further perform a systematic generalization study across countries and product types, from which we reveal an asymmetric generalization phenomenon: models trained on more liquid markets or products transfer well to less liquid ones, whereas the reverse transfer leads to substantial performance degradation.

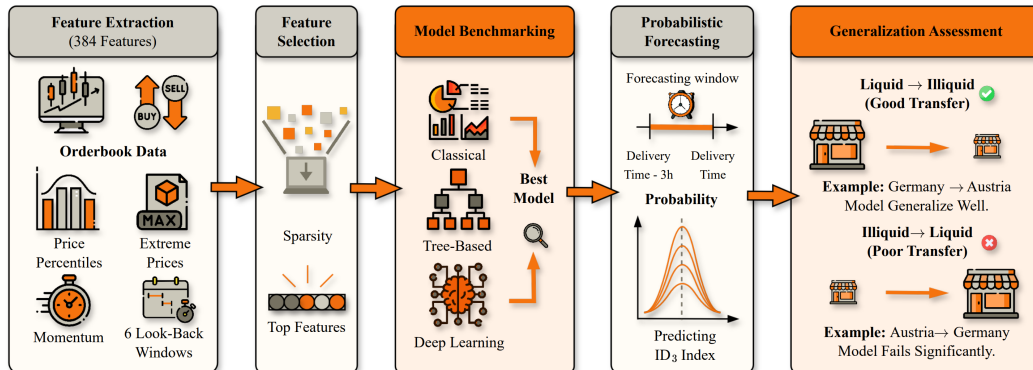


Figure 1: Workflow of Orderbook Feature Learning and Asymmetric Generalization.